

Data 250 Practice Exam 3 Solutions

Q1

1.

$$2x + 3y + 5z = 8$$

$$x - y + z = 0$$

a. Write this system of equations as a matrix equation.

b. Solve the system of equations for the general solution

A1a

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In[7]:= Row[{MatrixForm[{{2, 3, 5}}, {1, -1, 1}]],  
         " . ", MatrixForm[{x, y, z}], " = ", MatrixForm[{8, 0}]}
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$$\text{Out[7]} = \begin{pmatrix} 2 & 3 & 5 \\ 1 & -1 & 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 8 \\ 0 \end{pmatrix}$$

A1b

under-determined

$$x = 8/5 - 8z/5$$

$$y = 8/5 - 3z/5$$

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In[9]:= Reduce[{{2, 3, 5}}, {1, -1, 1}}.{x, y, z} == {8, 0}]
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$$\text{Out[9]} = y == \frac{8}{5} - \frac{3z}{5} \ \&\& \ x == \frac{8}{5} - \frac{8z}{5}$$

Q2

2. Given three $n \times n$ matrices $A = (a_{ij})$, $B = (b_{ij})$, $C = (c_{ij})$

Write the i -th row and j -th column of the matrix multiplication $A B C$ using summation notation

A2

Out[17]=

$$\sum_{k=1}^n \sum_{m=1}^n a_{ik} b_{km} c_{mj}$$

Q3

Solve for the matrix X given that $A X A^{-1} = Y$

A3

Carefully multiply the inverses to the other side.

Use that the inverse to A^{-1} is simply A which is multiplied on the right

$$A X = Y A$$

then A^{-1} on the left for both sides

$$X = A^{-1} Y A$$

Q4

Given that the 6×7 matrix X has its null space spanned by $[1, 3, -1, 0, 0, 2, 5]$

a) what is the rank of X ?

b) what is the dimension of the image of $T(v) = X v$?

c) Let $Y = X X^t$

Show that Y is a symmetric matrix.

d) Show that Y is invertible

A4a

The null space plus the rank is the number of columns

$$1 + 6 = 7$$

the rank = 6

A4b

The dimension of the image is the rank = 6

A4c

$$(AB)^t = B^t A^t$$

$$(A^t)^t = A$$

Put these identities together.

$$Y^t = (X X^t)^t = (X^t)^t X^t \text{ by the first one}$$

$$= X X^t \text{ by the second identity}$$

which is Y by definition so Y is symmetric.

A4d

Y is a square matrix of full rank so it is invertible

Q5

$$A = \begin{pmatrix} 2 & 3 \\ x & 6 \end{pmatrix}$$

a) For what values of x is A nonsingular?

b) Calculate A^{-1}

when $x = 1$

A5a

In[24]:= **Det**[{{2, 3}, {x, 6}}]

Out[24]=

$$12 - 3x$$

$x \neq 4$ iff A is nonsingular

A5b

In[25]:= **Inverse**[{{2, 3}, {1, 6}}]

Out[25]=

$$\left\{ \left\{ \frac{2}{3}, -\frac{1}{3} \right\}, \left\{ -\frac{1}{9}, \frac{2}{9} \right\} \right\}$$

Q6

Suppose $Av = 3v$ and $Au = 2u$ where $u = [1, 1]$ and $v = [1, 2]$

a) Let $e_1 = [1, 0]$ and $e_2 = [0, 1]$. Calculate Ae_1 and calculate Ae_2

b) Find the four entries a_{ij} of A

c) What are the eigenvalues of A

A6a

$$e_1 = 2u - v$$

$$e_2 = -u + v$$

$$Ae_1 = 4u - 3v = [1, -2]$$

$$Ae_2 = -2u + 3v = [1, 4]$$

A6b

The first column of A is Ae_1

The second column of A is Ae_2

$$\text{so } A = \begin{pmatrix} 1 & 1 \\ -2 & 4 \end{pmatrix}$$

In[26]:= **Eigensystem**[{{1, 1}, {-2, 4}}]

Out[26]=
 {{3, 2}, {{1, 2}, {1, 1}}}

A6c

The eigenvalues are 3 and 2

Q7

7. Suppose the singular values of A are 3, 2, 1. What is the best estimate for the norm of A v for a nonzero vector v

A7

3 ||v|| since 3 is the biggest stretching of A